

ATYRAU INTL, Kazakhstan

WMO: 381411

Lat: 47.122N

Lon: 51.821E

Elev: -22

StdP: 101.59

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-20.1	-26.1	0.3	-23.5	-22.8	0.5	-20.2	12.4	-2.6	11.4	-4.0	2.9	50	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	38.2	19.9	37.0	19.8	35.2	19.3	23.4	29.7	22.3	29.8	21.3	30.2	5.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	16.5	26.8	20.0	14.7	25.3	18.8	13.6	24.7	69.7	29.0	65.1	29.7	61.7	29.8	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.9	10.6	9.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-25.6	39.8	4.6	1.7	-28.9	41.0	-31.6	42.0	-34.1	43.0	-37.4	44.2		
			-25.8	25.9	4.5	1.6	-29.0	27.0	-31.6	28.0	-34.1	28.9	-37.4	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-7.8	-6.8	2.3	11.3	19.6	24.6	26.7	25.5	18.3	9.6	1.7	-3.5	(6)
(7)		DBStd	13.18	6.46	6.71	4.26	4.95	4.40	4.04	3.37	4.13	4.26	4.65	5.19	5.18	(7)
(8)		HDD10.0	2038	553	470	240	41	1	0	0	0	1	63	249	419	(8)
(9)		HDD18.3	3768	811	703	496	216	37	4	0	3	52	271	498	678	(9)
(10)		CDD10.0	2120	0	0	2	81	298	437	518	481	249	52	2	0	(10)
(11)		CDD18.3	809	0	0	0	6	75	191	260	225	51	1	0	0	(11)
(12)		CDH23.3	10911	0	0	1	69	979	2599	3616	3061	574	11	0	0	(12)
(13)		CDH26.7	5757	0	0	0	16	414	1392	2036	1708	190	1	0	0	(13)
(14)	Wind	WSAvg	4.9	5.1	5.2	5.6	5.5	5.2	4.8	4.6	4.4	4.6	4.6	4.5	5.1	(14)
(15)	Precipitation	PrecAvg	183	15	12	15	16	27	17	10	9	16	19	17		(15)
(16)		PrecMax	340	34	27	47	45	90	46	40	38	41	47	50	48	(16)
(17)		PrecMin	113	1	0	0	0	1	0	0	0	0	3	1		(17)
(18)		PrecStd	50	10	8	13	11	23	15	10	12	9	13	11	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	6.1	17.8	29.1	35.8	39.2	40.0	40.1	33.9	24.9	15.8	7.2	(19)
(20)			MCWB	3.5	3.7	10.1	15.5	17.2	18.9	20.1	19.6	17.9	14.2	12.0	5.6	(20)
(21)		2%	DB	2.2	3.9	14.0	25.1	33.2	37.2	38.1	38.2	30.9	22.1	13.0	5.9	(21)
(22)			MCWB	2.1	2.3	8.4	14.3	17.3	19.3	20.3	19.9	16.5	13.1	9.9	4.7	(22)
(23)		5%	DB	1.2	2.2	11.2	22.2	31.1	35.8	36.9	36.8	28.8	19.9	11.0	4.1	(23)
(24)			MCWB	1.1	1.3	6.9	12.6	16.6	19.0	20.2	19.6	16.1	12.5	8.7	3.4	(24)
(25)		10%	DB	0.2	1.1	9.0	20.0	29.0	33.8	35.1	34.8	26.9	17.2	9.0	2.8	(25)
(26)			MCWB	0.0	0.5	5.6	12.2	16.3	18.7	19.8	19.2	15.6	11.1	7.2	2.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	4.8	11.7	16.9	21.0	23.1	25.9	24.9	20.7	17.3	13.1	6.5	(27)
(28)			MCDWB	3.9	5.6	15.6	23.0	27.9	28.8	32.6	29.6	27.5	21.8	14.1	7.0	(28)
(29)		2%	WB	2.2	2.7	9.2	15.5	19.4	22.0	23.9	23.1	19.3	15.0	10.8	4.9	(29)
(30)			MCDWB	2.3	3.2	13.6	22.2	27.0	29.2	29.7	30.3	25.6	18.6	12.0	5.4	(30)
(31)		5%	WB	1.1	1.6	7.5	14.3	18.5	21.1	22.7	21.7	18.1	13.4	8.9	3.3	(31)
(32)			MCDWB	1.2	2.1	10.4	21.0	26.3	29.6	29.8	30.3	25.2	17.8	10.6	3.9	(32)
(33)		10%	WB	0.1	0.7	6.0	13.0	17.5	20.3	21.8	20.6	16.9	12.1	7.0	2.0	(33)
(34)			MCDWB	0.2	1.2	8.2	18.1	25.4	29.2	30.2	30.9	24.3	16.6	8.6	2.4	(34)
(35)	Mean Daily Temperature Range	MDBR	6.7	8.2	9.0	11.5	12.6	13.5	13.5	14.1	12.9	11.2	7.8	5.6	(35)	
(36)		5% DB	MCDBR	5.0	7.1	12.8	14.0	15.0	15.6	15.2	16.0	14.9	13.7	10.2	6.1	(36)
(37)			MCWBR	4.6	5.9	7.9	6.4	6.1	6.0	6.0	6.4	6.9	7.7	7.2	5.2	(37)
(38)		5% WB	MCDBR	4.9	6.3	11.1	12.4	12.6	13.1	13.1	13.7	13.2	11.6	8.8	5.3	(38)
(39)			MCWBR	4.6	5.6	7.4	6.2	6.1	6.5	6.6	7.3	7.5	7.6	6.9	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.358	0.376	0.434	0.496	0.480	0.454	0.455	0.460	0.417	0.398	0.372	0.369	(40)	
(41)		taud	2.053	2.007	2.007	1.920	2.030	2.164	2.184	2.139	2.202	2.184	2.167	2.077	(41)	
(42)		Ebn at Noon	670	752	769	759	792	816	809	785	784	727	658	602	(42)	
(43)		Edh at Noon	98	126	148	178	165	145	140	141	121	105	88	86	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.96	1.79	3.27	4.88	6.20	6.88	6.62	5.83	4.43	2.69	1.40	0.81	(44)	
(45)		RadStd	0.17	0.31	0.40	0.37	0.45	0.49	0.32	0.28	0.27	0.24	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+1.33	N/A	-0.73	N/A	N/A	+231	+155

Nomenclature: See separate page